Sekar M

Ph.D. Research Scholar
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RESEARCH INTERESTS

Natural Language Processing, Large Language Models, Reasoning for Language Models, Multi-Hop Question Answering, Retrieval-Augmented Generation,Question Answering, Mixture-of-Experts (MoE), Multilingual and Multi-modal AI

TECHNICAL SKILLS

- **Languages & Databases:** Python, C, C++, Java, Neo4j
- **Frameworks & Libraries:** PyTorch, HuggingFace (Transformers/PEFT), LangGraph, scikit-learn, NumPy, NLTK
- **Deep Learning & LLMs:** Fine-tuning (LoRA/PEFT), Mixture-of-Experts (MoE), Graph Neural Networks (GNNs)
- **Tools & Systems:** Linux, Git, LaTeX, Docker

EDUCATION

National Institute of Technology Tiruchirappalli, Tamil Nadu, India — July 2024 – Present
Ph.D. in Computer Science and Engineering — CGPA: 8.5/10

Government College of Engineering, Bargur (Anna University), Tamil Nadu, India — July 2022 – May 2024
M.E. in Computer Science and Engineering — CGPA: 8.0/10

Varuvan Vadivelan Institute of Technology (Anna University), Tamil Nadu, India — August 2018 – May 2022
B.E. in Computer Science and Engineering — CGPA: 7.5/10

JOURNAL PUBLICATIONS

- **Sekar M**, Avula Karthik, Santhanavijayan A. ***Do Reasoning Models Resist Knowledge Editing? Evidence from Edit Adherence in Multi-Hop Question Answering***. *Knowledge-Based Systems*, Elsevier (2026). Q1, SCIE indexed. (Under Review)

CONFERENCE PUBLICATIONS

Accepted / Published

- **Sekar M**, Pranava Mohan, Santhanavijayan A. ***MOEZHI-GURU: Multi-Perspective Misogynistic Meme Detection via Multimodal Mixture of Experts***. Multimodal Misogyny Detection Workshop (MMLoW) @ AACL-IJCNLP 2026. (Accepted)
- **Sekar M**, Jatavath Sirisha, Santhanavijayan A. ***TalkingToM: Dual-Process Cognitive Modeling for Narrative Theory of Mind Question Answering***. 5th International Conference on Speech and Language Technologies for Low-Resource Languages (SPELLL 2026), Amrita Vishwa Vidyapeetham, Coimbatore, December 17–19, 2026. (Accepted – Oral)
- **Sekar M**, Sirisha Jatavath, Santhanavijayan A. ***ViBrainNet: An Attention-Based CNN–Vision Transformer Ensemble for Brain Tumor Detection and Segmentation in MRI***. 2nd Congress on Intelligent Machines and Algorithms (CIMA 2026), NIT Puducherry, May 23–24, 2026. (Accepted)
- **Sekar M**, J. Nafeesa Begum, K. Kumar, S. Selvi. ***RowColumnPrune: An Efficient Compression of Large Language Models through Row and Column Deletion***. 10th International Conference on Latest Trends in Science, Engineering and Technology (ICLTSET'24), April 2024. ISBN 978-93-6228-897-4.
- **Sekar M**, Chandrakanth K, Sivasakthi R, Vasanthakumar M. ***A Machine Learning Approach to Dimensionality Reduction in Large Data sets***. National Conference on Computing & Information System (NCCIS'22), May 2022.

Under Review

- **Sekar M**, Saikiran M, Santhanavijayan A. ***Separating Retrieval from Reasoning: Diagnosing ReAct Agent Failures in Multi-Hop Question Answering***. International Conference on Natural Language Processing (ICON 2026). (Under Review)

- **Sekar M**, Sudarsan A, Adharsan N, Santhanavijayan A. ***GradeRAG: Retrieval-Augmented Question Answering That Remembers Its Mistakes***. Forum for Information Retrieval Evaluation (FIRE 2026), Kolkata, India, December 17–20, 2026. (Under Review)
- **Sekar M**, Buddha Vignesh Kumar, Chinikeri Harishwar Reddy, Santhanavijayan A. ***RRR: Reduced Tokens, Rapid Inference, Regressed Answers from Fine-Tuning-Time Vocabulary Extension in Telugu***. ACL Rolling Review (ARR), October 2026 cycle. (Under Review)
- **Sekar M**, Rudra Narayana Tangi, Santhanavijayan A. ***CrossTalk: Measuring and Mitigating Cross-Topic Context Bias in Multi-Turn LLMs***. ACL Rolling Review (ARR), October 2026 cycle. (Under Review)

ACADEMIC EXPERIENCE

Department of CSE, NIT Tiruchirappalli — Tiruchirappalli, Tamil Nadu, India
Ph.D. Research and Teaching Experience (HTRA) — Jul'24 – Present

- Teaching Assistant, **CSPC31 – Principles of Programming Languages** — Jul'26 – Present
- **Course Instructor**, **CSIR15 – Essentials of Computer Science** — Jan'26 – May'26
- Laboratory Teaching Assistant, **CSLR62 – Web Technology Laboratory** — Jan'26 – May'26
- **Course Instructor**, **CSHO12 – Distributed Systems** — Jun'25 – Dec'25

Capgemini — Bengaluru, Karnataka, India
Software Developer Intern — Feb'22 – Jun'22

CONFERENCES & WORKSHOPS ATTENDED

Sarvam Epoch – Builder Edition — Bengaluru, India
Sarvam AI — 29th July 2026

6th Indian Symposium on Machine Learning (IndoML 2025) — BITS Pilani Hyderabad, India
Graduate Forum – Poster Presentation: *TamilaAR: A Tamil Analogy Test Set for Multilingual Analogical Reasoning in Large Language Models* — Accepted

ACM Annual Event — Coimbatore, India
PSG College of Technology — 1st March 2025

21st International Conference on Natural Language Processing (ICON 2024) — Chennai, India
NLP Association of India (NLPAI) – AU-KBC Research Centre — 19th – 22nd December 2024

Lecture: How Could Machines Reach Human-Level Intelligence? — Chennai, India
Speaker: Yann LeCun (VP & Chief AI Scientist, Meta) – Indian Institute of Technology Madras — 22nd October 2024

Build with AI — Bengaluru, India
Google Developer Groups (GDG) — 23rd March 2024

Art of Computing Workshop — Bengaluru, India
Department of CSA, Indian Institute of Science — 17th – 18th February 2024

ACHIEVEMENT

- **Overall 2nd Rank** in the **Cross-Cultural Misogynistic Meme Detection (CC-MMD 2026) Grand Challenge**, co-located with the ACM International Conference on Multimodal Interaction (ICMI 2026).